

OM RAJENDRA KATHALKAR

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PROFESSIONAL SUMMARY

PhD researcher in Computer Science at the National Technical University of Athens, focusing on efficient and deployable AI systems, particularly inference optimization under real-world constraints. My work spans computer vision, edge AI, and network-aware systems through academic research and industry collaborations. Alongside research, I am deeply interested in teaching and mentoring, and enjoy helping learners build strong foundations in AI through clear, practical guidance.

EDUCATION

PhD in Computer Science,

National Technical University of Athens (NTUA), Greece

Fall 2025

Advisor: [Dr. Konstantinos Tserpes](#)

Master of Science in Electronics and Communication Engineering,

International Institute Information Technology (IIIT-H) - Hyderabad

Advisor: [Dr. Sachin Chaudhari](#)

Thesis: [Camera-Based Deep Learning Framework for AQI Estimation: Dataset and Methodology](#)

Bachelor of Engineering in Electronics and Telecommunication Engineering,

St. Vincent Pallotti College of Engineering and Technology - Nagpur

RESEARCH EXPERIENCE

Research Associate

Jan 2026 - Present

[INSTITUTE OF COMMUNICATION AND COMPUTER SYSTEMS \(ICCS\), NTUA, Greece](#)

- Conducting research on **efficient and autonomous execution of AI models**, with focus on **edge inference, and scalable AI systems** in collaboration with NTUA research groups.
- Contributing to European research initiatives aligned with **Horizon Europe** projects, working on AI-driven optimization, real-world deployment constraints, and reproducible research pipelines.
- Collaborating with interdisciplinary teams across academia and industry to translate theoretical models into **deployable AI solutions** for connected systems.

Applied AI Researcher

Dec 2024 - Dec 2025

[ERICSSON RESEARCH, BANGALORE](#)

- Contributing to a European Project called [PANDORA](#) on network-aware path planning for autonomous mobile robots (AMRs).
- Developing algorithms that combine real-time decision-making and semantic understanding for reliable navigation in dynamic industrial environments keeping the network constraints in equation.

Research Scholar

Aug 2021 - Oct 2025

[Signal Processing and Communications Research Centre, IIIT-H](#)

- Pursuing research with [Dr. Sachin Chaudhari](#) on the edge AI for IoT and Computer Vision towards a smarter Internet of Things.
- Led development of a sensor-free AQI estimation framework using camera data, reducing dependence on costly, failure-prone sensor networks and created TRAQID (26,678+ traffic-related images with co-located weather data), presented as a Spotlight Paper at ICVGIP 2024, and among the most downloaded papers of the conference.
- Proposed AQIFormer, a transformer-based multi-view model achieving 90% AQI classification accuracy, including 88% at night and 81.67% in cross-city transfer, demonstrating strong generalization.

TEACHING EXPERIENCE

- Guiding machine learning, deep learning, and NLP concepts to industry professionals with courses from IIIT Hyderabad, IISc Bangalore and the University of Michigan.
- Educated and mentored **250+ industry professionals** across diverse domains on **Agentic AI and Generative AI**, engaging learners with **15+ years of industry experience** and enabling practical upskilling through hands-on workflows, real-world use cases, and applied AI systems.

AI Instructor

April 2025 - January 2026

iHUB-DATA, HYDERABAD

- Delivering structured modules on classification, regression, clustering, PCA, neural networks, and ML/DL applications in a 24-week AI/ML student training program.
- Facilitated hands-on tutorials, practical evaluations, and performance tracking for more than 50+ engineering students.

Instructor for Embedded Systems Workshop

Monsoon 2024

Instructor for ML for IoT Workshop

Monsoon 2023

Instructor for Machine Learning

KEIT, Spring 2022

PUBLICATIONS

Journal Papers

[J1] **Om Kathalkar**, Houssam Hajj Hassan, Ajay Kattepur, Georgios Bouloukakis, “Network-aware path planning for AMRs: A generalizable framework with spatial-aware adaptation”, *Autonomous Robots (Springer Nature)*, under review.

Conference Papers

[C5] **Om Kathalkar**, N. Nilesh, S. Chaudhari, A. Namboodiri, “**AQIFormer: A Transformer-Based Multi-View Architecture for Cross-City Air Quality Classification**”, *Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, 2024.

[C4] **Om Kathalkar**, Houssam Hajj Hassan, Ajay Kattepur, Georgios Bouloukakis, “**Network-Aware Path Planning for Autonomous Mobile Robots in Industrial Environments**”, *IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Delhi, 2025. (Acceptance rate: 32.8%)

[C3] **Om Kathalkar**, N. Nilesh, S. Chaudhari, A. Namboodiri, “**TRAQID - Traffic-Related Air Quality Image Dataset**”, *Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, Bangalore, 2024. **(Spotlight Presentation)** 

[C2] Spanddhana Sara, Andrew Rebeiro-Hargrave, **Om Kathalkar**, Sachin Chaudhari, “**Protocol for Hunting PM2.5 Emission Hot Spots in Cities**”, *1st International Workshop on Advances in Environmental Sensing Systems for Smart Cities (EnvSys)*, Helsinki, Finland, 2023. 

[C1] K.S. Viswanadh, **Om Kathalkar**, Sachin Chaudhari, “**CV and IoT-based Remote Triggered Labs: Use Case of Conservation of Mechanical Energy**”, *FiCloud*, Rome, Italy, 2022. 

Patents

[P3] Indian Patent Filed: “System for Determining Air Quality Index (AQI) in Urban Environments”, Application No. 202541073518, May 2025.

[P2] Indian Patent Published: “**System and Method for Determining Air Quality by Processing Environmental and Traffic-Related Visual Data**”, Publication No. 202441043706, May 2024.

[P1] US Patent Published: “**System and Method for Implementing an Experiment Remotely and Determining an Output Using a Computer Vision Model**”, Publication No. US20240078641A1, March 2023.

RESEARCH PROJECTS

Remote Labs Platform

2021 – 2022

Funding: PRIF & Department of Science and Technology (DST), India

Institution: IIIT-Hyderabad

Developed and supported a large-scale remote laboratory platform to **democratize STEM education**, enabling access to experimental learning for **thousands of rural and underrepresented students** across India.

IoT Enabled Smart Cities: Pollution, Health and Governance

2019 – 2023

Funding: PRIF & Department of Science and Technology (DST), India

Institution: IIIT-Hyderabad

Designed **image-based air quality monitoring** methods integrating traffic visuals and environmental cues to support **scalable, low-cost AQI estimation** for smart city planning and public health governance.

PANDORA (Horizon Europe Project)

2024 – 2025

Funding: Horizon Europe

Research on **network-aware and trustworthy AI** for Autonomous Mobile Robots (AMRs), focusing on **robust path planning, communication-aware decision-making**, and reliable operation in industrial environments.

TURING (Horizon Europe Project)

2026 – Present

Funding: Horizon Europe

Working on **efficient and autonomous execution of AI algorithms**, with emphasis on **scalable inference, trustworthy AI, and deployable AI pipelines** for real-world industrial systems.

SKILLS

Areas of Interest	Machine & Deep Learning, Edge AI, Internet of Things, NLP, Agentic & Gen AI
Tools	Pytorch, Tensorflow, Nvidia Sionna, Cuda, Django, Scikit-learn, L ^A T _E X, N8N
Hardware	Jetson Xavier NX, Raspberry Pi, Arduino and Esp-Series Micro-controllers.

LETTERS OF RECOMMENDATION

- **Dr. Sachin Chaudhari**, Associate Professor, IIIT Hyderabad, India. 
- **Dr. Ajay Kattapur**, Senior Researcher, Ericsson Research, Bangalore, India. 
- **Dr. Georgios Bouloukakis**, Assistant Professor, University of Patras, Greece. 
- **Dr. Houssam Hajj Hassan**, Senior Researcher, Orange Research, France. 

AWARDS AND HONORS

1. Awarded the **iHub-Data Research Fellowship** (August 2023) for MS by Research at IIIT Hyderabad; fellowship fully covered research funding and tuition fees based on academic merit and research potential.
2. **Winner** of the **International Environmental Sensing Project Competition** (March 2023), University of Helsinki, Finland; secured top position among teams from multiple participating countries.
3. Awarded the **AIR-1 Gold Award** at the **e-Yantra Ideas Competition** (IIT Bombay), a national-level competition featuring 100+ top engineering college teams across India.
4. Conferred the **Student of the Year** award (May 2023) among a batch of 1000+ undergraduate students at SVPCET, Nagpur, for outstanding academic performance, leadership, and overall contributions.
5. Received the **Best Project Award** (December 2022) for an innovative and technically rigorous undergraduate final-year project demonstrating novel problem-solving and research-oriented execution.
6. Honored as **Student of the Department (ETC)**, SVPCET, Nagpur, in recognition of consistent research contributions, academic excellence, and departmental engagement during undergraduate studies.